EXPERIMENT 6

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// Define time range and sampling interval t_min = -5;

 // Start time t_max = 5;

// End time dt = 0.01;

 // Time step t = t_min:dt:t_max;

 // Time vector

// Define Signal 1: Rectangular pulse of width 2 centered at t = 0 x_t = (t >= -1) .* (t <= 1);

// Define Signal 2: Exponential decay function h_t = exp(-t) .* (t >= 0); // Exponential decay, causal

// Perform the convolution y_t = conv(x_t, h_t) * dt;

// Multiply by dt to scale correctly

// Adjust the time vector for the convolution result t_conv = (2 * t_min):dt:(2 * t_max);

// Plot Signal 1 clf;

 // Clear the current figure

 subplot(3,1,1);

 plot(t, x_t, 'b'); xlabel('t');

 ylabel('x(t)');

 title('Signal 1: Rectangular Pulse');

// Plot Signal 2

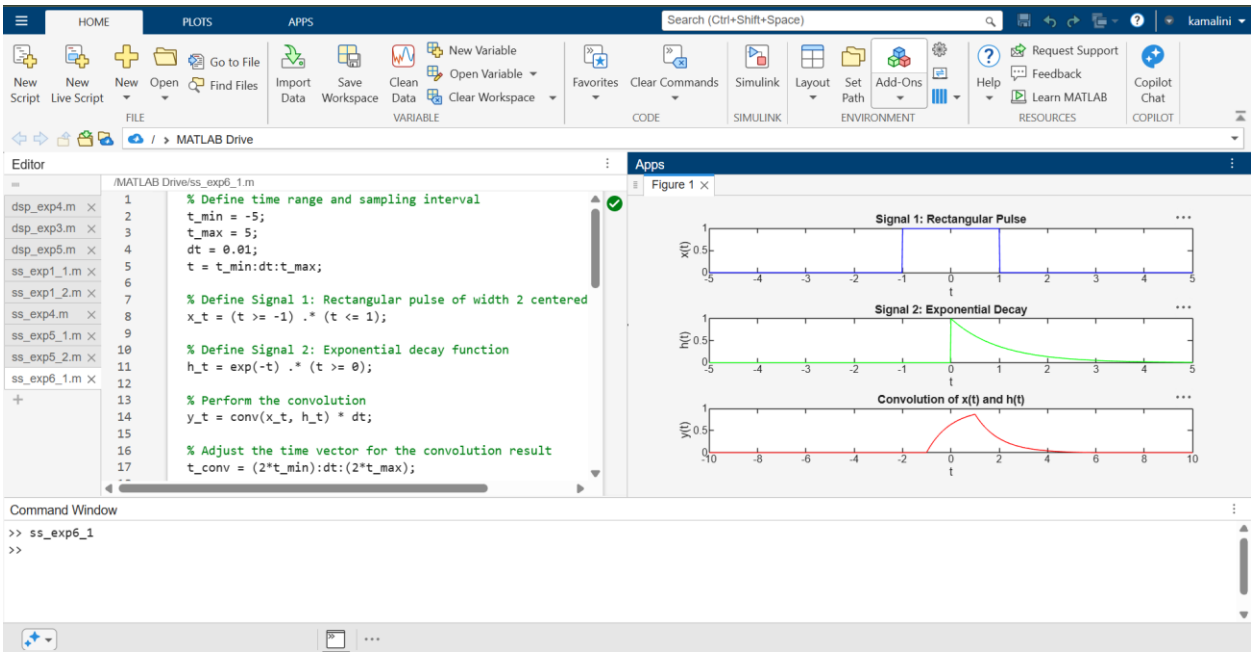
 subplot(3,1,2);

 plot(t, h_t, 'g');

xlabel('t'); ylabel('h(t)'); title('Signal 2: Exponential Decay');

// Plot the Convolution Result subplot(3,1,3);
```

plot(t_conv, y_t, 'r'); xlabel('t'); ylabel('y(t)'); title('Convolution of x(t) and h(t)');



2:

// Aliasing Effect Demonstration in Scilab clc; clear;

/ 1. Continuous-time signal (reference)

t = 0:1e-4:0.05;

// High resolution time vector x = sin(2*%pi*300*t) + 0.8*sin(2*%pi*700*t);

2. Sampling parameters

fs = 800; // Sampling frequency (Hz) Ts = 1/fs;

// Sampling period

// Sampled time vector n = 0:Ts:0.05; xs = sin(2*%pi*300*n) + 0.8*sin(2*%pi*700*n);

3. FFT of sampled signal

N = 2048;

// FFT length Xf = abs(fft(xs, N));

// Forward FFT, zero-padded to N f = (0:N-1)*(fs/N); // Frequency axis

4. Plotting

clf;

Continuous vs Sampled signal

subplot(2,1,1); plot(t, x, 'b');

plot(n, xs, 'ro'); legend("Continuous Signal", "Sampled Signal");

xtitle("Continuous vs Sampled Signal", "Time (s)", "Amplitude");

Spectrum of sampled signal subplot(2,1,2); plot(f, Xf, 'b');

a = gca(); a.data_bounds = [0, 0; fs/2, max(Xf)];

xtitle("Spectrum of Sampled Signal (Aliasing Effect)", "Frequency (Hz)", "Magnitude");

